Wi-Fi Standards, Architecture & Security Assignment

Comprehensive Long-Form Technical Report & Detailed Solutions (Tasks 1 to 4)

Task 1: Comparison of Popular Wi-Fi Standards (Speed & Frequency Bands)

Wireless networking standards defined by the IEEE 802.11 working group have evolved significantly over the decades to deliver higher data throughput, lower latency, and enhanced multi-device efficiency. Below is a detailed breakdown of three popular Wi-Fi standards:

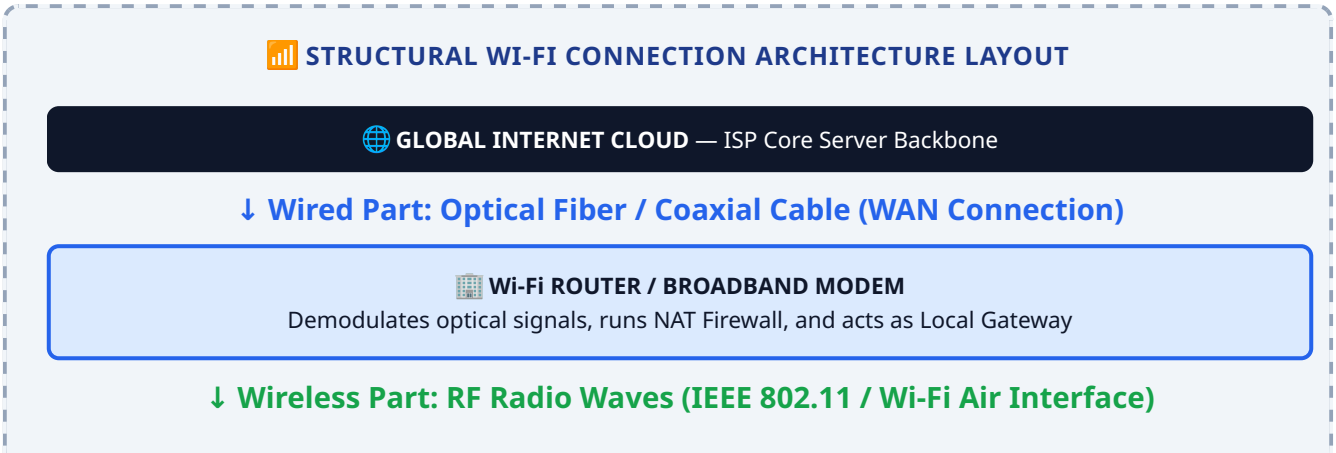
Wi-Fi Standard	Common Commercial Name	Supported Frequency Bands	Maximum Theoretical Speed
802.11n	Wi-Fi 4 (Released 2009)	2.4 GHz and 5 GHz (Dual-Band)	Up to 600 Mbps
802.11ac	Wi-Fi 5 (Released 2014)	5 GHz band exclusively	Up to 3.5 Gbps (3,500 Mbps)
802.11ax	Wi-Fi 6 / 6E (Released 2019)	2.4 GHz, 5 GHz, and 6 GHz (Tri-Band)	Up to 9.6 Gbps

Key Technical Differences:

- **802.11n (Wi-Fi 4):** Introduced MIMO (Multiple-Input Multiple-Output) technology, utilizing multiple antennas simultaneously to increase data rates and reach speeds up to 600 Mbps across dual frequencies.
- **802.11ac (Wi-Fi 5):** Shifted focus entirely to the wider, less congested 5 GHz frequency band, supporting wider channel bandwidths (up to 160 MHz) and multi-user MIMO (MU-MIMO) to achieve gigabit-plus speeds.
- **802.11ax (Wi-Fi 6):** Brought revolutionary efficiency enhancements using OFDMA (Orthogonal Frequency-Division Multiple Access) and introduced the brand-new 6 GHz band (Wi-Fi 6E), pushing speeds up to 9.6 Gbps in high-density environments.

Task 2: Smartphone-to-Internet Wi-Fi Connection Architecture Diagram

A home wireless setup bridges mobile wireless clients to the global internet by combining wireless RF radio transmissions with wired optical fiber/copper infrastructure. Below is the structured network diagram illustrating this connection:



📱 **Smartphone (Client Device)**
Sends/receives data packets via built-in Wi-Fi antenna

Explanation of Network Segments:

- **The Wireless Part (Air Interface):** The connection between your smartphone and the Wi-Fi router is entirely wireless. Data is modulated onto radio frequency (RF) waves ($2.4 ext{ GHz}$ or $5 ext{ GHz}$) through the air using IEEE 802.11 standards.
- **The Wired Part (Infrastructure Backhaul):** The Wi-Fi router connects to the outside world via a physical wired line (optical fiber or coaxial cable) provided by your Internet Service Provider (ISP), transmitting digital packets across the global internet backbone.

Task 3: Wireless Network Scan, Observation & Security Evaluation

Scanning the surrounding airwaves using a laptop or smartphone reveals multiple active Service Set Identifiers (SSIDs). Below is a documented sample scan log of three nearby Wi-Fi networks, along with an analytical decision on which network to select.

SSID (Network Name)	Signal Strength (RSSI)	Security Protocol	Security Evaluation & Status
1. Airtel_Fiber_5G_Home	-45 dBm (Excellent / Strong)	**WPA3-Personal**	Maximum encryption standard; highly secure against offline dictionary attacks.
2. Jয়ের_Free_Café_Wi-Fi	-68 dBm (Moderate / Fair)	**Open (No Security)**	Unencrypted public network; vulnerable to packet sniffing and man-in-the-middle attacks.
3. Neighbor_TP-Link_2.4	-82 dBm (Weak / Poor)	**WPA2-PSK (AES)**	Industry-standard secure protocol, but distant signal causes severe packet loss and slow speeds.

🎯 **Network Selection & Justification:**
• **Selected Network:** `Airtel_Fiber_5G_Home`
• **Reasoning:** When choosing a network, both **signal strength** and **security** are paramount. `Airtel_Fiber_5G_Home` offers an exceptional signal strength (-45 dBm) ensuring high-speed throughput, coupled with the state-of-the-art **WPA3** security protocol, providing robust cryptographic protection against unauthorized access. The open public network is rejected due to severe security risks, while the weak neighbor network is avoided due to poor signal quality.

Task 4: Comprehensive Analysis of 'SSID Hiding' (Pros & Cons)

What is SSID Hiding?
Every wireless router periodically broadcasts its network name (SSID) in management frames known as *Beacons* so that nearby devices can discover it. Enabling **"SSID Hiding"** (or disabling SSID broadcast) instructs the router to stop including the network name in these beacon frames. Consequently, the network appears as "Hidden

Network" or blank in device Wi-Fi lists, requiring users to manually type both the exact SSID name and password to connect.

Advantages of Hiding Your Wi-Fi SSID:

- **Reduced Casual Visibility:** It hides your network name from casual neighbors, visitors, or passersby scanning for open or available Wi-Fi networks on their phones, reducing unwanted connection attempts.

Disadvantages of Hiding Your Wi-Fi SSID:

- **False Sense of Security & Privacy Leaks:** SSID hiding does *not* encrypt your traffic or prevent determined hackers from discovering your network. In fact, hidden networks are actually *more* prone to passive sniffing because your connected devices continually broadcast probe requests containing your hidden SSID name into the air looking for your router, allowing malicious actors to easily capture the real SSID name in plain text! Furthermore, it causes connection friction for legitimate guests.